allowed this; but for a majority of the SEP heads, the ‘compromise’ stemmed from their belief, which sooner than later turned out to be false, that the Indian scientists and engineers cannot build a rocket even if they are allowed to work here for five years. In other words, we were taken for granted as a bunch of inefficient men masquerading as scientists to have a long, paid vacation in France. We were just too eager to prove the French wrong, and we did it in a few months, after which the Indian scientists were acknowledged as some of the best SEP had seen and worked with.

This we achieved through hard and systematic toil. To begin with, I drew up a ten-phase programme, which would take us from scratch to rocket engine.

The ten phases were roughly as follows:

Phase 1: Identification of subsystems and sub-subsystems, breaking them down to the last, nut, bolt, pin and clip

Phase 2: Specifications, with tolerance levels of each subsystem

Phase 3: Detailed drawings

Phase 4: Materials, nomenclature and specifications

Phase 5. Fabrication technology

Phase 6: Non-conformances and acceptance criteria

Phase 7: Detailed design report

Phase 8: Fabrication and assembly

Phase 9: Cold tests and hot runs

Phase 10: Integration of the rocket engine into the stage

I insisted on deploying Indian scientists in such a way that they would mix with their French counterparts and not remain segregated. Some were not very happy with us not sitting together. I explained to them that integrating ourselves with the French would help us learn the language quicker. What I would tell them only a little later was that this would enable us to learn not just French, but the French way of making rocket engines.

Once we broke down the project, I put one man in charge of each subsystem. This took us three months. Every evening at six we assembled to discuss what was learned that day. Once a week, we also met to listen to team leaders talk about what they tried but could not learn, and why. I was